

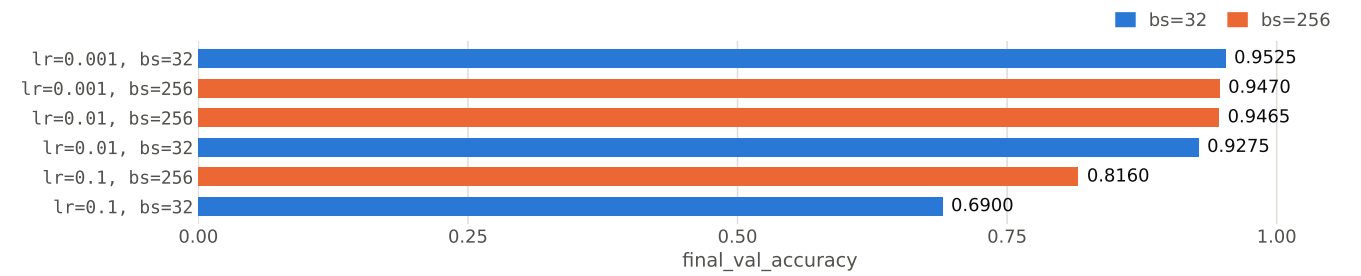
1. Results

lr	bs	final_train_loss	final_val_acc	best_val_acc	overfit_gap
0.001	32	0.0053	0.9525	0.9545	0.0463
0.001	256	0.0231	0.9470	0.9470	0.0492
0.01	256	0.0288	0.9465	0.9530	0.0440
0.01	32	0.0932	0.9275	0.9420	0.0466
0.1	256	0.5387	0.8160	0.8795	0.0244
0.1	32	1.0876	0.6900	0.8010	0.0136

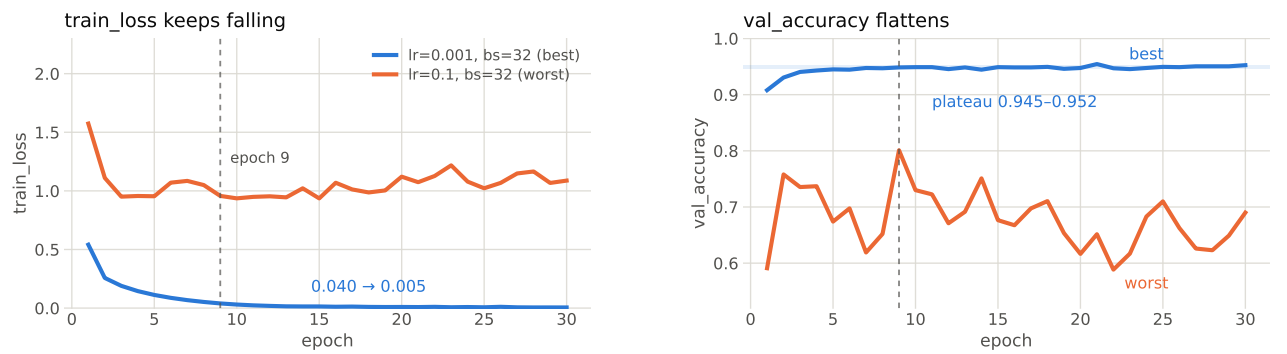
Fixed across all runs: 128 hidden units, 30 epochs, Adam, seed 42, 10,000 train / 2,000 validation.

2. Analysis

Best-Performing Run learning_rate=0.001 with batch_size=32 won, at final_val_accuracy 0.9525 (best 0.9545) and the lowest final_train_loss, 0.0053. The step size is small enough for Adam to stay stable on this MLP yet large enough to converge inside 30 epochs — training loss falls monotonically, 0.541 to 0.005. At lr=0.1 the step overshoots and the run never settles (loss 1.088, accuracy 0.690) and lr=0.01 sits in between. The small batch helps at the margin: 10,000 samples at bs=32 gives 313 updates per epoch against 40 at bs=256, so more progress per epoch, while the extra gradient noise stays mild at this step size. All six runs ranked below, coloured by batch size.



Overfitting Evidence The curves for the winning run, train_loss keeps falling after roughly epoch 8 — 0.040 down to 0.005, an eightfold drop — while val_accuracy flattens in the 0.945–0.952 band and merely wobbles (0.9545 at epoch 20, back to 0.9475 by epoch 23). Final train accuracy is 0.9988 against 0.9525 validation: an overfit_gap of 0.0463, with lr=0.001/bs=256 marginally worse at 0.0492. The extra epochs buy memorisation, not generalisation — below, the winning run in blue against the worst run in orange, with the plateau band shaded.



Larger Effect Effect of learning-rate effect is **0.2625** (max – min over all six runs) against a effect of batch-size **0.0502** (mean |Δ| within each learning-rate pair). Learning rate dominates by 5.2×. The batch-size gap is negligible at lr=0.001 (0.0055) and only matters once the run is already unstable (0.1260 at lr=0.1). The MLflow parallel-coordinates view below confirms it: across all six runs the lines fan out over the learning_rate axis and collapse together over batch_size.

